

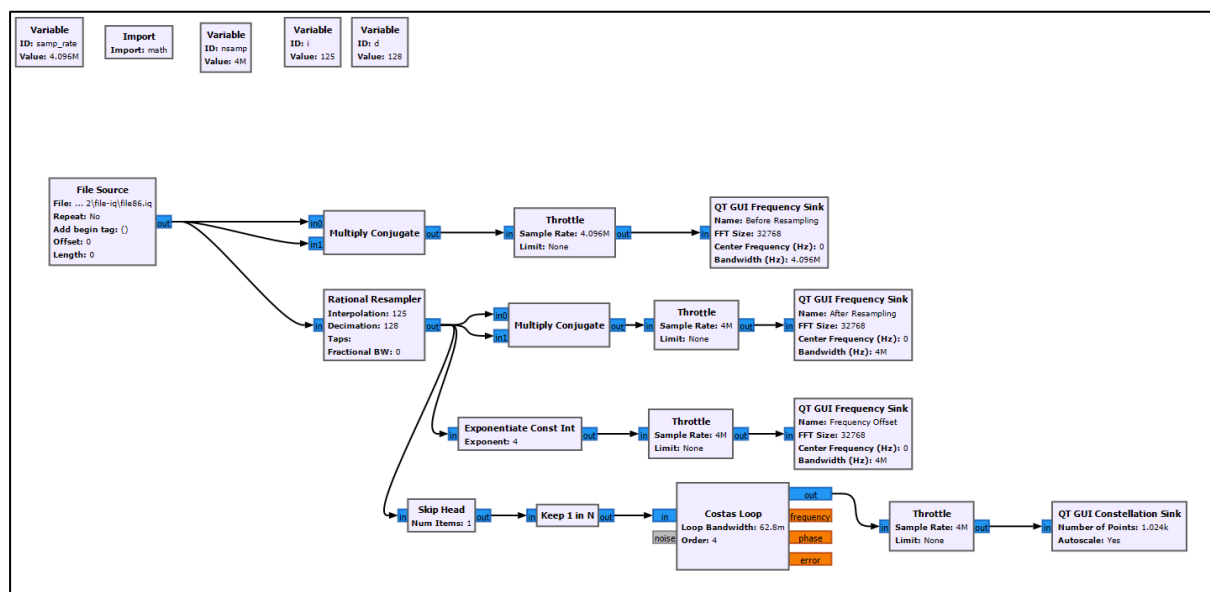
EE 332 Digital Communication

Project

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220102094

Flow Diagram :

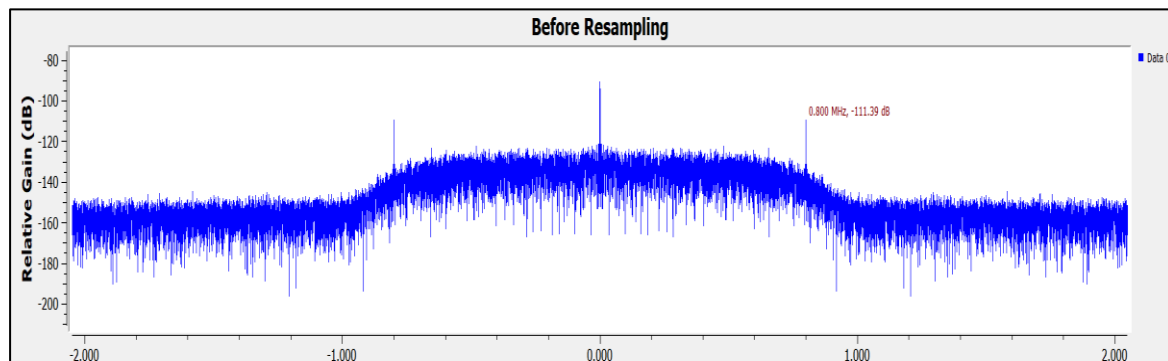


Techniques Used :

- 1] Obtaining the symbol rate : Multiplied the signal $r(nT_s)$ with its conjugate $r^*(nT_s)$ and using FFT sink obtained peaks at intervals of $1/T$ from centre frequency (0Hz).
- 2] Resampling : As I was getting fractional sample per symbols, I had to resample my data to a new frequency.
- 3] Obtaining constellation scheme : Used Costa's Loop for carrier recovery while checking for the sample with minimum ISI.

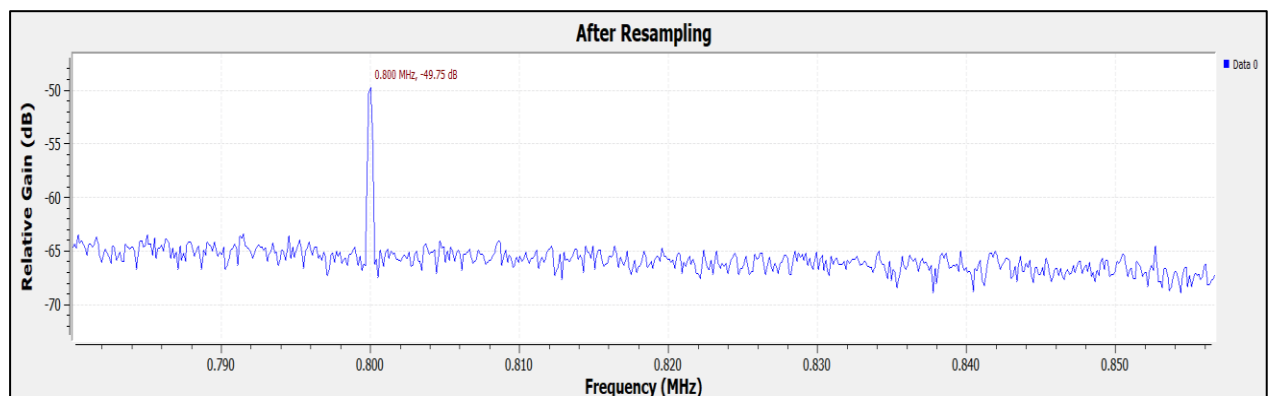
4] Obtaining frequency offset : Exponentiated the signal according to the constellation scheme and took the FFT to observe shift in the original spectrum and the exponentiated spectrum.

Calculations :

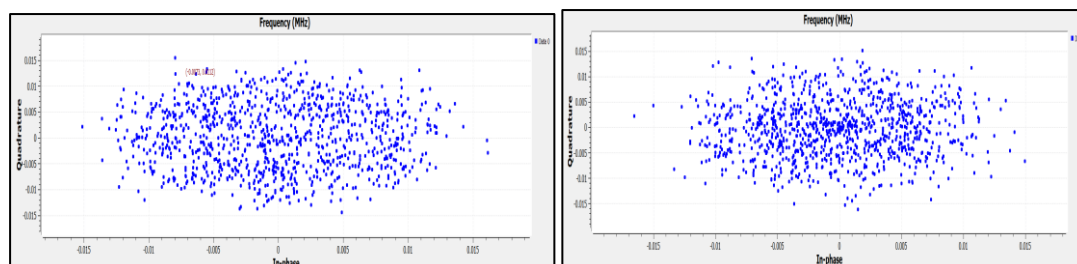


The peaks were obtained at 0.8MHz and hence the symbol rate=0.8MHz. Here, sampling frequency $F_s=4.096\text{MHz}$.

Samples per symbol= $4.096/0.8=5.12$ which is fractional.

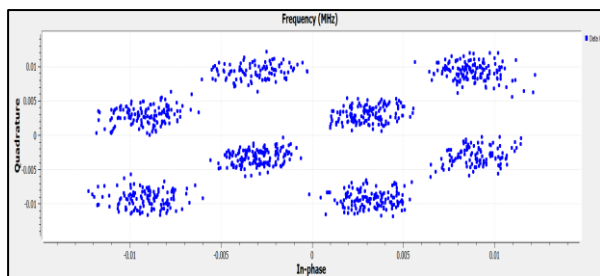


So, lowering F_s to 4MHz gave 5 samples per symbol.

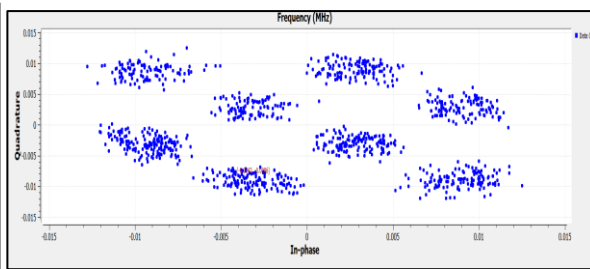


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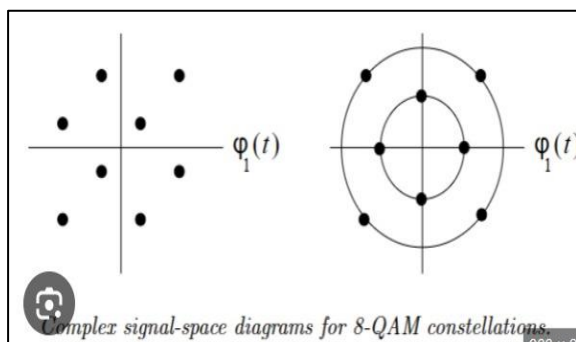
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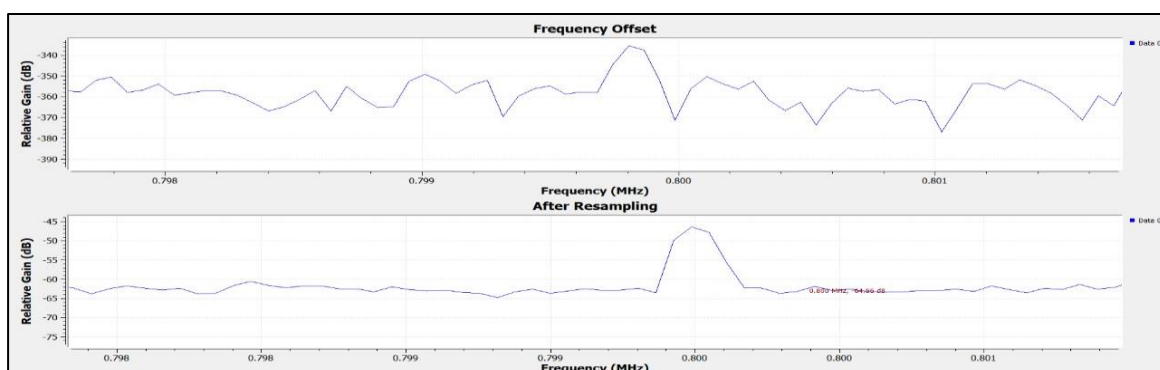
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Then, using Costa's Loop the constellation scheme was obtained as 8-QAM.



Now, keeping the exponent value as 8, the frequency shift was obtained as $0.8 - 0.7996 = 0.0004 \text{ MHz} = 400 \text{ Hz}$.

Hence, $q\Delta f = 400$ and $q = 8$. Therefore, $\Delta f = 50 \text{ Hz}$.

Results:

Symbol Rate	0.8MHz=800kHz
Samples per symbol	5
Constellation Scheme	8-QAM
Frequency Offset	50Hz

